

Muhammad Hazrat AI Engineer

USA | +1 (484) 581-9691 | muhammadhazrat621@gmail.com | [LinkedIn](#)

SUMMARY

AI/ML Engineer with 2.5+ years of experience building **AI automation, computer vision, deep learning, and client-facing AI solutions**. Progressed from award-winning sign-language research and **machine learning model development** to computer vision applications and **AI-powered workflow automation**. Delivered **95% dispatch automation** and a **91%-accurate sign-language model** through technical ownership, stakeholder collaboration, model evaluation, and practical problem solving.

SKILLS

Programming & Databases: Python, Java, SQL

AI & ML: LLMs, RAG, NLP, DL, Prompt Engineering, Computer Vision, LSTM, Image Processing, Pose Estimation, Model Evaluation

Frameworks & Libraries: TensorFlow, Keras, PyTorch, OpenCV, MediaPipe Holistic

Backend & Data Systems: REST APIs, FastAPI, Flask, Data Pipelines, Apache Kafka, gRPC, Distributed Systems, Scalable Systems

Cloud & DevOps: Amazon Web Services (AWS), Google Cloud Platform (GCP), Kubernetes, Jenkins, Git, GitHub, Linux, CI/CD

AI Integration & Automation: Microsoft 365 Copilot, Microsoft Power Automate, Microsoft Teams, ChatGPT, Gemini, Hatz.ai, ConnectWise PSA API, PIA.ai APIs, CloudRadial

Methodologies: Agile, Scrum, Requirements Refinement, Acceptance Criteria, Prototype Validation, Iterative Testing, User-Feedback Analysis, End-User Training

EXPERIENCE

AI Integration Engineer & IT Technician | BizTech Helpers

May 2025 – Present | USA

- Automate 95% of previously manual IT dispatch steps by integrating **ConnectWise PSA API, Microsoft Power Automate, and Microsoft Teams**, enabling real-time alerts, AI-assisted ticket triage, and cross-client coordination.
- Implement **Microsoft 365 Copilot** and **Hatz.ai** across client workflows, automating repetitive tasks and delivering individual and group training that improves AI adoption and documented workflow efficiency by 30%.
- Integrate **PIA.ai APIs** into managed service provider operations, standardizing AI-assisted ticket handling and collaborating with technicians to reduce resolution time by 25% across client accounts.
- Develop **CloudRadial** portal solutions and LLM-based support workflows using **ChatGPT** and **Gemini**, improving ticket submission, client communication, troubleshooting consistency, and support-process efficiency by 40%.

Machine Learning Engineer Intern | VentureDive Ltd.

Jun 2024 – Sep 2024 | USA

- Engineered a **computer vision virtual try-on pipeline** using **image preprocessing, prototyping, and product visualization workflows**, increasing product-trial engagement by 30% through testing with engineering and product stakeholders.
- Optimized virtual try-on quality through **image-processing algorithm research, visual evaluation, and computer vision tuning**, improving user satisfaction by 20% across four months of experimentation, reviews, and prototype refinements.
- Developed reusable **computer vision workflows** for ML experimentation, model evaluation, prototype validation, and product visualization, improving testing consistency and reproducibility across engineering reviews.
- Collaborated with engineering and product stakeholders to refine virtual try-on requirements, prioritize model improvements, and convert user feedback into acceptance criteria and validated prototypes.

Machine Learning Capstone Researcher | Forman Christian College University

Aug 2022 – Aug 2023 | Pakistan

- Engineered **HandAI** a real-time **Pakistan Sign Language translation application**, applying **machine learning, dataset curation, and data preprocessing** to achieve 91% recognition accuracy and improve accessibility for deaf users.
- Designed a **deep learning pipeline** using **MediaPipe Holistic, TensorFlow, Keras, and LSTM neural networks**, extracting hand, facial, and pose keypoints to achieve 91% sequence-based gesture recognition accuracy across input modalities.
- Optimized **real-time inference** by converting trained models to **TensorFlow Lite** and integrating a **Flask REST API**, enabling **on-device predictions** and supporting **mobile deployment** throughout the 12-month development and testing cycle.
- Drove 12 months of **product iteration, cross-functional collaboration, and user-feedback analysis**, working with faculty and stakeholders to prioritize feature improvements and secure two innovation awards: APICTA and PASHA for HandAI.

PROJECTS

Personalized Customer Support System | LangFlow, GPT-4o Mini, RAG, Python

- Developed an AI-powered customer support system using **LangFlow, GPT-4o Mini, RAG, and Python**, delivering context-aware responses and improving consistency across complex customer inquiries.
- Built scalable retrieval and **backend orchestration** workflows using structured data, prompt engineering, and **LLM generation**, strengthening information retrieval, answer accuracy, and reusable AI components.

HandAI 2.0 | Kotlin, Android Studio, TensorFlow Lite, Python, Flask, MediaPipe

- Developed **HandAI 2.0**, a real-time Android sign-language translation application using **Kotlin, Android Studio, TensorFlow Lite**, and computer vision, achieving 91% accuracy with offline accessibility.
- Integrated **MediaPipe, Python, and Flask REST APIs** to process gesture keypoints and optimize low-latency on-device inference, earning Drexel University recognition for technical innovation and accessibility.

EDUCATION & CERTIFICATIONS

Master of Artificial Intelligence and Machine Learning,

Sep 2023 – Mar 2025

Drexel University, Pennsylvania, USA | GPA: 3.81/4.00

Machine Learning Specialization | Stanford University [[Link](#)]

Data Science Specialization | IBM [[Link](#)]

Jul 2022

Sep 2023